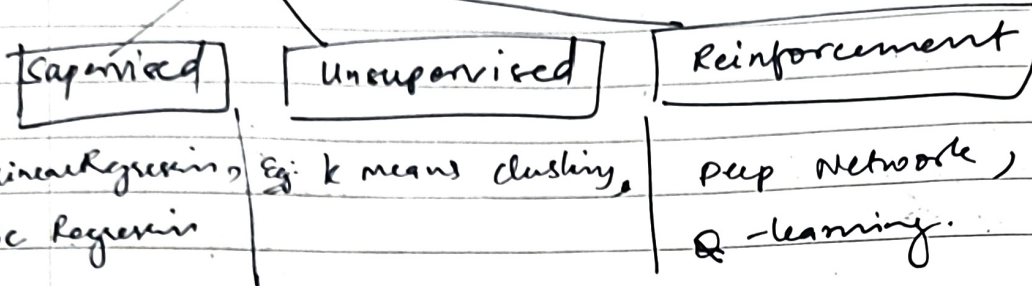


### ① Types of ML with examples:



### ② Feasibility of learning :-

\* An project planning tool.

\* It asks: "How well a model ~~performs~~ performs  
~~for~~ "Can a ML algorithm solve a given  
problem efficiently when given enough resources  
and time?"

Key factors:

\* Data Availability

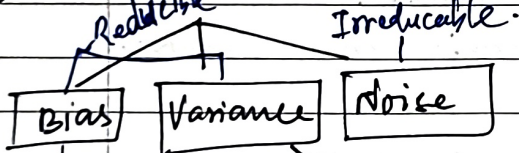
\* Computational resources

\* Problem complexity

\* Nature of problem

→ checking feasibility = Higher chance of success.

### ③ Types of Errors



Caused to bestfit  
model that are  
not centred towards (0,0).

### Types of Noise

Measure  
ment Error

Random  
fluctuations.

Error from  
something  
to small fluctuations

### ④

| Bias                                                                                                                           | Variance                                                                                               | Tradeoff:                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>* underfits</li> <li>* model too simple &amp; can't learn important patterns</li> </ul> | <ul style="list-style-type: none"> <li>* overfitting</li> <li>* model learns noise too much</li> </ul> | <p>↑ model complexity<br/>= Bias ↓, Variance ↑</p> <p>↓ model complexity<br/>= Bias ↑, Variance ↓</p> |

⑥ Hypothesis space : possible solution

\* means "A possible solution".

DATE 

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

\* Represented by  $|H|$

\* Set of all possible model an algorithm can choose to solve a problem.

\* Types:

Finite Hypothesis  
 $|H| < \infty$

Infinite Hypothesis  
 $|H| = \infty$

\* no. of possible models are finite

\* no. of possible models are infinite

⑦

dataset

Training  
80%

Testing  
20%

Accuracy:

Measures how well a model predicts on unseen data.  
i.e., test data.

⑧

Approximation :

\* How close <sup>or</sup> a model can fit on the data is called Approximation.

\* If a model is too simple, it can't identify the true relationships in data, which is called Approximation error.

Generalization :

How well a model performs on unseen data or test data. If the model learns too well, even learning the noise.

Tradeoff :

Model complexity  $\uparrow$  = Approximation Error  $\downarrow$  Generalization Error  $\uparrow$

Model complexity  $\downarrow$  = Approximation Error  $\uparrow$  Generalization Error  $\downarrow$

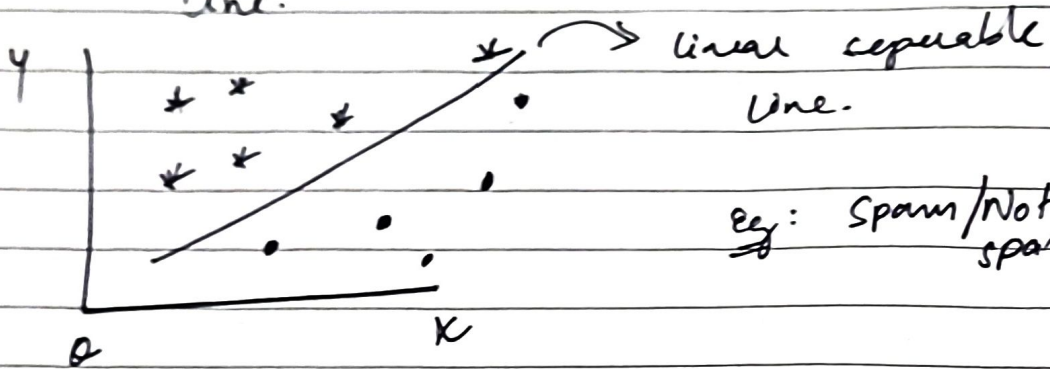
⑨

Underfitting, Overfitting  $\approx$  : Already known.

## ① Classification types :

### \* Linear classification : or classes

→ Data points can be separated by drawing a single separable line.

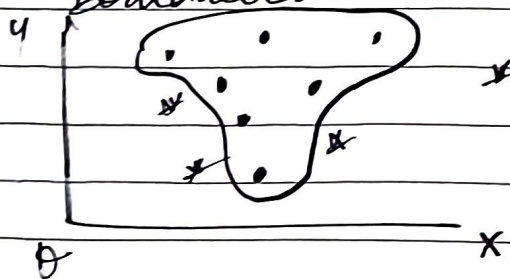


eg: Spam/Not spam.

### \* Non-linear classification :

\* The classes are not linearly separable

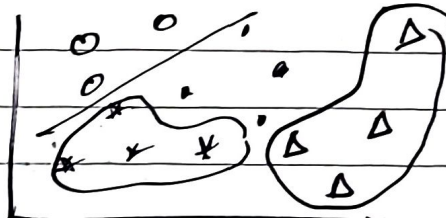
\* But can be separated by boundaries.



eg: Decision tree, neural network.

### \* Multiclass Classification :

\* It has two or more classes.



\* It could be linear / non-linear.

### \* Multilabel Classification

\* Each sample can have multiple classes

eg: movie genres (Action, Comedy, Sci-fi, etc)

## VC dimension :

- \* max. no. of separable points over the hypothesis space  $H$
- \* If VC dimension  $\uparrow$ , model flexibility  $\uparrow$  but also a chance of overfitting.
- \* If VC dimension  $\downarrow$ , model is simple but may lead to underfit
- \* Represented by  $VC(H)$ .

## Decision Trees :-

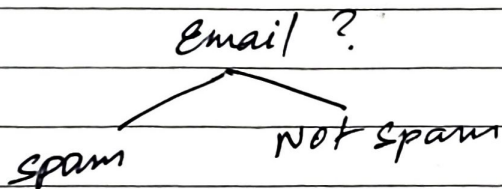
[Refer pre-mids pdf]

Classification Tree

Regression Tree

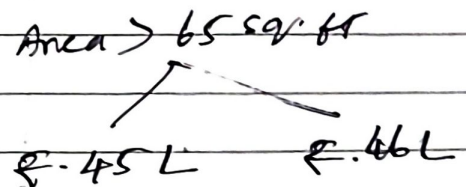
used to classify data into categories.  
(discrete values)

eg: Email is  
spam / Not spam



used to predict continuous values (numerical)

eg: to predict  
= house price based  
on features like location  
and size



## ⑤ Numericals — Decision trees.

⑥ Entropy :  
— measure of amount of randomness in a data.

$$\text{Entropy} = - \sum_{i=1}^c P_i \log_2 (P_i)$$

$$\underline{Gini} = 1 - \sum_{i=1}^c P_i^2$$

↓

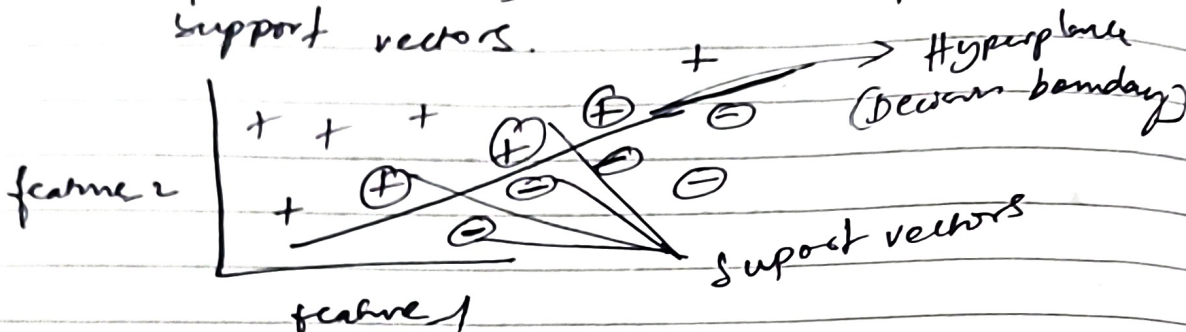
↑  
proportion  
of instances  
of class  $i$

probability of randomly choosing an element in a dataset.

7

SVM :

- \* A supervised ML algorithm used for classification tasks & Regression tasks.
- \* It tries to find a Hyperplane that separates the datapoints of different classes with help of support vectors.



Types

- Linear SVM
- Non Linear SVM
- SVM for Regression.

8 KNN

9 ~~Regression~~

Numericals.

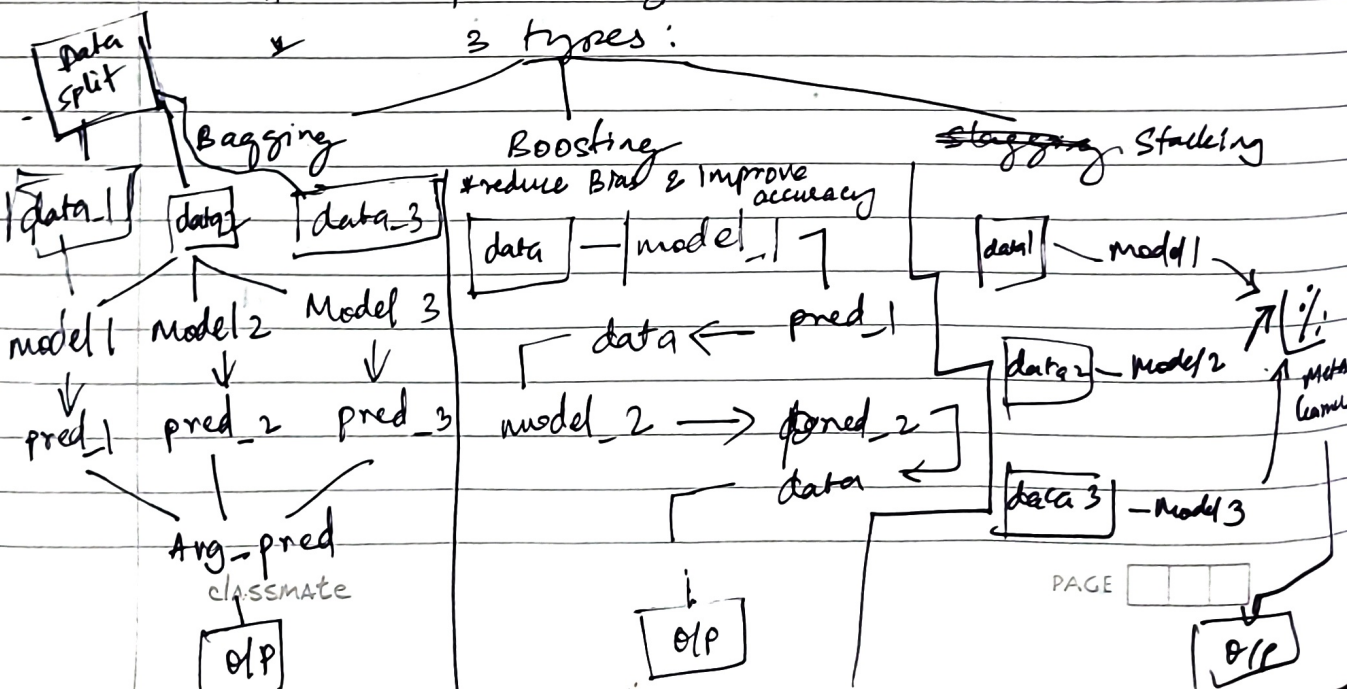
10 Explain CART:

- \* Classification & Regression Trees.
- \* Uses Gini Impurity.
- \* Minimizes Error.

### Module - 3

1 Ensemble Learning :

- \* Combines multiple models to solve a single problem.
- \* Improves generalization.
- \* 3 types :



K means clustering numericals.  
K means, K mode (refer pdf).

PCA: → An example of Dimensionality Reduction.

\* Principle Component Analysis:

\* A method in ML; reduces the no. of features in data while keeping the most important information.

— to reduce noise / unwanted information

— makes models faster.

\* Reduces no. of variables in a data.

⑤ ECOC: (Diagram ???)

\* Error correcting output codes.

\* comes under Ensemble learning technique.

\* Used to decompose multi-class classification into multiple binary classification to handle classification errors more effectively.

\* Consists of two steps:

1. Encoding

2. Decoding

⑥ Voting Schemes: (comes under Ensemble learning)

\* Combines predictions with multiple model to improve accuracy & robustness.

\* Types:

Hard voting

Each model makes its prediction. The class ~~that~~ (the prediction) that gets highest votes wins.

Soft voting

Takes average of predictions and gives the output

Weighted voting

The models with high accuracy will be more weighted in voting. Their

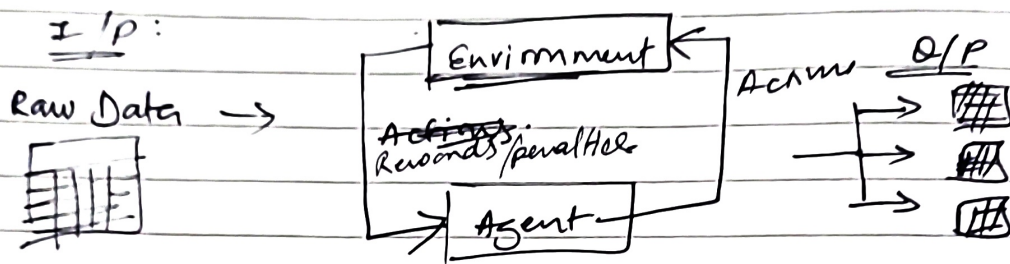
votes count more in final decision.

# Module - 4

## ① Reinforcement learning:

→ Learns by rewards & penalties  
Components: (Working principle)

- \* Agent: Decision maker (Learner)
- \* Environment: Where the agent is located.
- \* State: The condition / situation the agent is in.
- \* Goal: The state that Agent wants to reach.
- \* Action: Choices the agent makes



## ② Active learning vs Passive learning:

③

models actively & selectively  
learns from a ~~fixed~~ dataset  
& also queries <sup>for specific labels</sup> to improve  
learning efficiency reduces labelling  
costs.

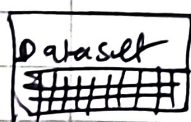
eg: Asking a teacher for  
questions to learn fast.

eg: Medical Diagnostics,  
Fraud detection, NLP tasks.

Model learns from  
a fixed datasets. It  
doesn't strive to  
improve by querying.  
It simply trains on  
what it has

eg: learning from  
a fully written  
book.

eg: Image classification.



→ Active learner

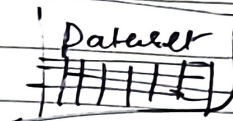
Select  
samples (orally  
Labels)

↓  
Add labelled  
data

make  
prediction

Model  
Trained.

classmate



→ Passive  
Learner

Trained model

↓  
Make  
prediction.

Exploitation:

- \* using best actions that are known already
- \* optimize current performance based on existing knowledge.
- + Immediate & Reliable rewards.

DATE

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

⑦ → Dynamic Programming:

- \* method used in Reinforcement learning to solve problems by breaking it into smaller ones, solving them recursively using Bellman Equation.

→ Direct utility estimation:

Every time our agent visits a state, it adds up all future rewards from that point. Then, it averages those totals across many visits to estimate how good the state is.

→ Temporal Difference Learning:

- \* model-free method in Reinforcement learning that updates value estimates after every step, using difference b/w current and next state predictions.

Module-5

⇒ Refer ANN.